Scalable and Open AI Solutions: From Cloud to Sovereign to Edge

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OCP GLOBAL SUMMIT

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AI is entering a new operating era



Scalable compute for distributed environments

Optimized performance for storage & networking

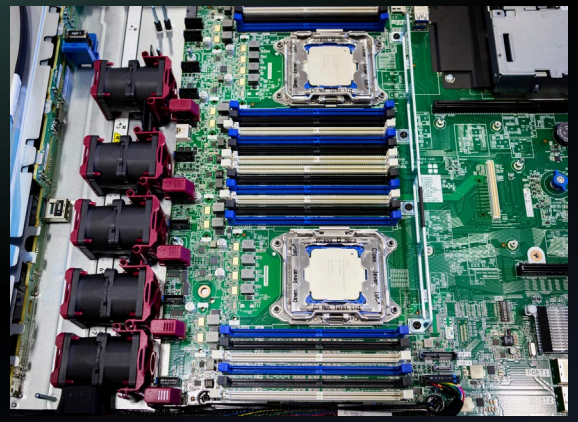
Highest security protection

Maximum operational efficiency

Heterogeneity is the future of AI



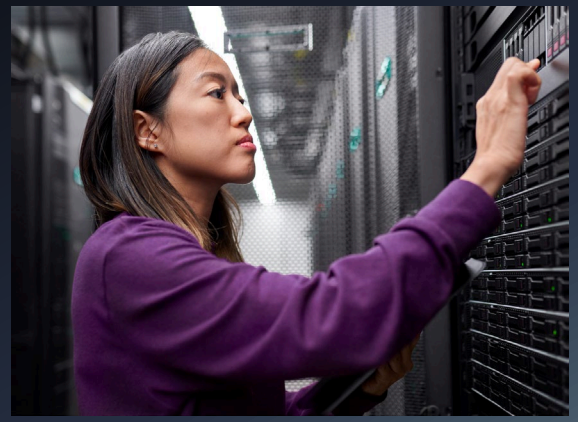
Performance across distributed environments



Multiple processors and architectures



Reimagined silicon, memory and rack-scale designs



Easy to connect, deploy, and operate

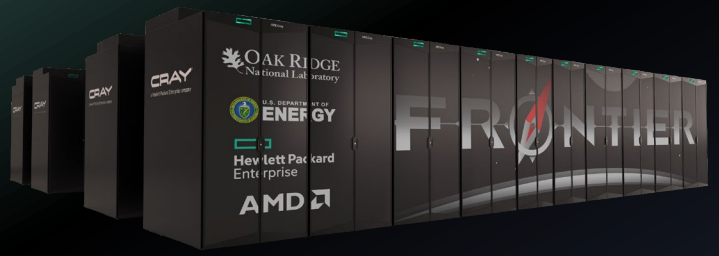
Decades of leadership innovation powers a new AI era

HPE has built 6 of the **world's top 10 supercomputers**

#2



#3



#4



#1 system with highest AI performance

13 of top 25 most energy-efficient supercomputers

300+ patents in liquid-cooling technology

Pioneer in open networking

First national AI system for Genesis Mission with open Ethernet

Industry leader chip-to-cloud security

OCP matters more than ever in AI



Open at every layer

AI requires full stack heterogeneous open technologies

Open compute



Open networking



Open systems



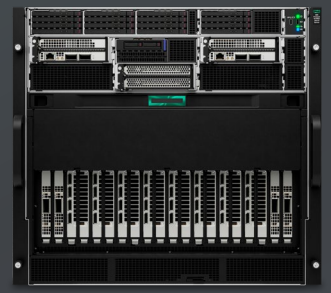
HPE and OCP collaborate to drive open architecture for AI

Advancing modular, secure, AI-accelerated open compute from edge to cloud

Open specifications based modular hardware systems (MHS)



HPE ProLiant DL525 Gen13

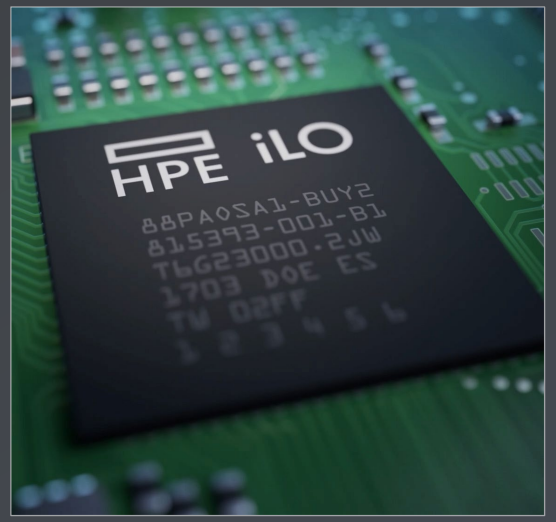


HPE ProLiant DL585a Gen13

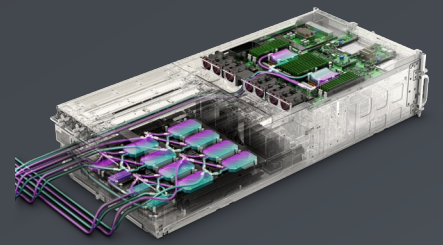


HPE ProLiant XD2300 Gen13

Secure by design, from chip-to-cloud



Engineered for optimal efficiency



Direct liquid-cooling

Advancing open networking through open collaboration

Broadening AI Data Center adoption with Ethernet-based scale-up, scale-out, and scale-across

Choice of Network OS

SONiC

Junos

Compliance (Open Industry Specifications)

Ultra Ethernet Consortium

ESUN

Ethernet for Scale-Up Networking

Support for all Use Cases (Scale-up, Scale-out, Scale-across)



QFX Switches

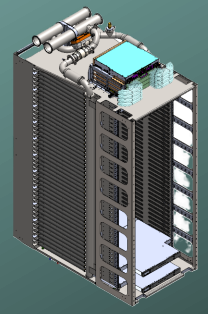


PTX/MX Routers

Comprehensive, Scalable and Self-driving Networking for AI Data Centers

Advancing open systems for large-scale AI with the OCP community

OPEN SYSTEMS TECHNOLOGIES

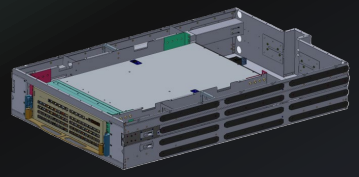


HPE Cray Supercomputing GX5000

OCP-inspired rack infrastructure for upcoming supercomputers

HPE Slingshot networking moving to UEC-based

Support for non-HPE networking



Support for OCP Components

Liquid cooling for OCP NIC 3.0 and leading contributor to specification

Industry-first DLC ORv3 QFX 1.6T switch

Open technologies deployed at scale to address next-gen scientific discovery

LARGE-SCALE CUSTOMER SYSTEMS



Mission

Los Alamos NATIONAL LABORATORY



Vision

Los Alamos NATIONAL LABORATORY



Discovery

OAK RIDGE National Laboratory

OCP partners collaborate to drive new directions in leadership-class AI systems

“For decades, AMD and HPE have ushered in breakthrough engineering together, including the world’s first exascale supercomputer. We continue to build on our joint innovation and looking forward to powering the next generation of exascale and AI machines with open technologies for systems like Discovery and Lux for the Department of Energy’s Oak Ridge National Laboratory.” – Thomas Zacharia, Senior Vice President, Global Public Sector and Strategic Partnerships, AMD

HPE | AMD | OAK RIDGE National Laboratory



OCP partners collaborate to drive new directions in leadership-class AI systems

"Scientific computing increasingly requires AI and simulation to work together at scale. We're co-designing Mission and Vision with HPE and Los Alamos to meet that need. The systems will combine NVIDIA Vera CPUs and Rubin GPUs with HPE's supercomputing architecture and support for OCP server designs, accelerating complex simulations and AI-driven discovery." – *Ian Buck, Vice President and General Manager, Hyperscale and HPC, NVIDIA*

HPE | NVIDIA | Los Alamos NATIONAL LABORATORY



Take the next step

Collaborate with us on OCP contributions to usher in a new era of AI

Open compute



HPE ProLiant DL340 Gen12

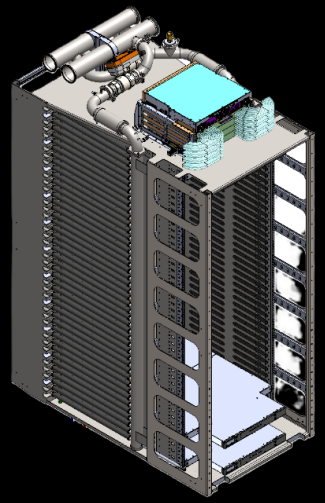


HPE ProLiant DL525 Gen13

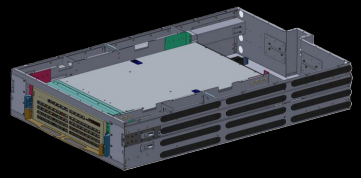


HPE ProLiant XD245 Gen13

Open AI systems



HPE Cray Supercomputing GX5000



OCP Adapter

Open networking



QFX Switches



PTX/MX Routers

AI-DC Demo : HPE Networking Solutions

SONiC Demo : Reimagined to deploy DCs

AI Grid Demo: HPE AI Grid Solution